

Valeriy Onuchin

Senior Software Engineer

Protvino, Moscow region, Russia

Application and systems programming, scientific and medical software, DevOps.

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ABOUT

Almost 40 years in research institutes and in industry. I know what physicists need from software, and how to build it so it keeps running for years.

Co-author of ROOT, CERN's data analysis framework, which was used in the discovery of the Higgs boson. Twelve years of software for a proton therapy system. Most recently, Linux distributions and operating systems for x86_64, Elbrus and aarch64.

ROOT
Co-author: GUI, graphics, signals and slots, about 520 commits

12 years
Proton therapy software, more than 1,000 patients

h-index 43
15,108 citations in Scopus

EXPERIENCE

03.2025 – 12.2025

Software Engineer · ELVEES R&D Center

Altel OS, based on ALT Linux, for the Russian Elbrus-2000 and Skif (aarch64) platforms; cross-compilation.

- OS images for the Skif SoC (1892VA018, 4 × Cortex-A53): 8 variants, from build host and router to GNOME and KDE, Linux kernel 6.6.

mkimage-profiles-skif ALT image build profiles for Skif.

2022 – 2024

Systems Programmer · JINR, Dubna

Software for the SPD experiment at the NICA collider, Joint Institute for Nuclear Research.

- Docker image for distributed computing: detector simulation and physics event reconstruction (Geant4, Pythia8, ROOT, Gaudi).
- Automated from-source build of a Linux distribution: 170+ packages, including maths and physics libraries, for x86_64, e2k and aarch64.
- CI/CD pipelines for automated Docker image builds and testing.
- CMake infrastructure for a heterogeneous development environment.
- Co-author of the SPD detector Technical Design Report (2024).

lfspd Docker image for parallel computing, based on Linux From Scratch.
SpdCMake CMake functions for the SPD experiment (internal JINR repository).

2009 – 2021

Lead Medical Software Developer · Protom, Protvino

Software for proton cancer therapy: a 3D DICOM viewer, a patient video monitoring system (C#, C++, real time), diagnostics, databases. The Protom synchrotron runs in the Prometheus and Radiance 330 systems in Protvino, Obninsk, the USA and Israel; by 2021 they had treated more than 1,000 patients.

SOFTWARE

3D DICOM Viewer	3D CT viewer with contour and dose editing; converts DICOM into the treatment planning system's format. C#, C++, DevExpress.
AlignPatient	Patient position monitoring: live video is compared with a reference image; video and audio link to the doctor, pulse and breathing recording. C#, C++, COM, DirectShow; multicast UDP, TCP/IP, RTP, SIP.
PFS/TM	C# implementation of the data exchange protocol inside the accelerator complex; also works as a PowerShell module.
MainCClient.NET	Client that sends the treatment plan to the server and starts, pauses or cancels irradiation.
EBTfilm	Analysis of EBT2/EBT3 dosimetry films after proton beam exposure. C#, C++, ROOT.

2004 – 2008

Software Developer, ROOT team · CERN, Geneva

Co-author of ROOT, the data analysis framework of high energy physics: more than 4 million lines of C++, hundreds of thousands of users. More than 2 exabytes of data are stored in ROOT files today, and the Higgs boson was discovered with ROOT. About 520 commits to ROOT in 2000–2009.

MAIN CONTRIBUTIONS

- Windows GUI and graphics port of ROOT (TGWin32).
- The signals and slots mechanism (TQObject, TQConnection).
- GUI classes and the GUI Builder.
- Image processing classes, TASImage on top of libAfterImage (examples).
- The TGHtml HTML widget.
- Carrot, an Apache module that runs C++ scripts on the server side, like PHP.
- QtRoot (Qt, C++).
- RDBC (ROOT DataBase Connectivity): database access over ODBC, the JDBC API implemented as ROOT classes.

1987 – 2004

Research Scientist, Experimental Physicist · IHEP, Protvino

- Heavy-ion physics and quark-gluon plasma: co-author of key PHENIX collaboration papers at RHIC (USA).
- Neutrino physics: the MINOS experiment (Fermilab, USA).
- Kaon physics and rare decays: the Tagged Neutrino Facility (IHEP).
- The Hyperon-M experiment (U-70): a multi-computer data acquisition system.
- Design and operation of electromagnetic and hadronic calorimeters.

SKILLS

Languages	**C++**, **C#**, **Python**, Go, Kotlin, bash, PowerShell
Build	CMake, Meson/Ninja, autotools, Make, msbuild, pkg-config
DevOps	Docker, CI/CD, cross-compilation, Linux kernel, OS builds
Platforms	Linux, Windows, Android, Elbrus-2000, Skif (aarch64)
Expertise	High-load systems, cross-platform builds, scientific data analysis and visualisation, medical software.
Spoken	Russian: native. English: conversational; technical documentation.

PERSONAL PROJECTS

mtapi	Android HTTP bridge: 60+ API endpoints, debug overlay, browser.
ROOT.NET	C# to ROOT bridge: a ROOT canvas inside a .NET form.

PUBLICATIONS

43	15,108	78	34
h-index	citations	publications	in Physical Review Letters

Scopus (h-index, citations, publications) and ORCID (PRL papers). Scopus affiliation: JINR, Dubna.

SELECTED WORK

2009 **ROOT — A C++ framework for petabyte data storage, statistical analysis and visualization**
Antcheva I. et al. Comput. Phys. Commun. 180 (2009) 2499–2512 · 421 citations. Update for version 5.28: CPC 182 (2011) 1384, 126 citations.

2008 **The ALICE experiment at the CERN LHC**
ALICE Collaboration. JINST 3 (2008) S08002 · 1,996 citations

2008 **The magnetized steel and scintillator calorimeters of the MINOS experiment**
MINOS Collaboration. Nucl. Instrum. Meth. A (2008) · 294 citations

2007 **A multicomputer data acquisition complex based on MISS and SUMMA electronics for the Hyperon-M experiment**
Instrum. Exp. Tech. 50 (2007)

2024 **Technical Design Report of the Spin Physics Detector at NICA**
SPD Collaboration. arXiv:2404.08317

2020 **40 лет анализа больших объёмов данных в Физике Высоких Энергий: интервью с Рене Браном**
Habr, in Russian: an interview with René Brun, the creator of ROOT.

EDUCATION

1981 – 1987 **V. N. Karazin Kharkiv National University**
Faculty of Physics and Technology. Major: experimental nuclear physics.

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